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Jury Attention

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Jury Attention

Gary Nan Tie, Sep 10, 2025

Abstract

To address the Rashomon effect when multiple key-value semantics are present, we introduce a jury attention mechanism via fibrations. This accomplishes query updates that take into account differing world views.

Introduction

The iconic 1950 Japanese film *Rashomon*, directed by Akira Kurosawa, explores the nature of truth and perception. The term "Rashomon effect," has come to refer to situations which present multiple plausible yet conflicting perspectives on the same event. Differing world views are arrived at from different experiences. The complexity of exceptions necessitates a coherent systematic technique to unravel and resolve ambiguity.

Attention, the ability to focus on specific information while ignoring other distractions, is a crucial cognitive process that helps us learn, remember, and effectively perform tasks. This paradigm has been adopted by transformers in large language models, mathematically voiced as a scaled dot product, a form of statistical correlation. This database retrieval mechanism takes queries and tries to best match them to keys labeling values, iterating until perplexity is acceptable. Curiously the emergent behavior of transformer attention is the ability to often pass the Turing test despite natural language semantics being ignored. Fibration abductive attention on the other hand, learns a lifting to simultaneously update a query and find a key to best explain the value being returned, thereby taking language semantics into account.

A jury trial, or trial by jury, is a legal proceeding in which a jury makes a decision or findings of fact. This process enables rigorous debate to arrive at a consensus view and make a verdict. Analogously, we propose 'Jury Attention', via fibrations, to address the Rashomon effect when multiple key-value semantics are present. While our motivation is clear, its mathematical explication is arduous to intuit, so remember metaphorically "Pole pole!", the Swahili phrase meaning "slowly, slowly," commonly used by guides on Mount Kilimanjaro to encourage climbers to ascend at a slow pace to prevent altitude sickness. Patience, the summit will be worth it when you see how query updates can take into account differing key-value semantics.

Transformer attention mechanisms have many interesting architectures with features like being multi-headed, contextual, or abductive. Now there is one that is just! What will a jury of our peers look like in the future? Perhaps an application of jury attention is conflict resolution. In order for dialogue between two (or more) aggrieved parties to occur common ground must be found. Imagine each party training their own LLM on data that reflects their world view, thereby creating a key-value semantics embodying their values (pun unintended). Now use jury attention to build an oracle LLM that each party could consult in order to understand one another and explore if common ground could be found. Perhaps even previews of Solomonic verdicts could emerge! The oracle is mediating dialogue to facilitate self resolution of disputes. Protocols and safeguards for a fair, transparent and diplomatic oracle will need to be established. In general, for situations in which we want to make a decision that navigates competing interests, jury attention LLMs potentially have a role.

Jury Attention: query updates that take into account differing world views

Tokens representing words are projected into a lower dimensional latent space $\mathcal{L}$.

In $\mathcal{L}$, let subsets Q , K and V denote queries, keys and values respectively, and let Q' be a copy of Q , denote updated queries.

Data samples:

$$q_i \in Q, \quad i \in [n] = \{1, 2, \dots, n\}$$

$$\text{For } j \in J \geq 2, \quad (k_l^j, v_l) \in K \times V, \quad l \in [m]$$

We use this data to learn neural networks satisfying certain properties TBD.

Let $L: \mathcal{L} \times \mathcal{L} \rightarrow \mathbb{R}^+$ be a loss function.

For $\varepsilon > 0$, define $x \approx_\varepsilon y$ iff $L(x, y) < \varepsilon$.

Suppose there exists $\varepsilon > 0$ so that for $j \in \mathcal{J}$

we can learn neural networks $p_j: K \rightarrow V$

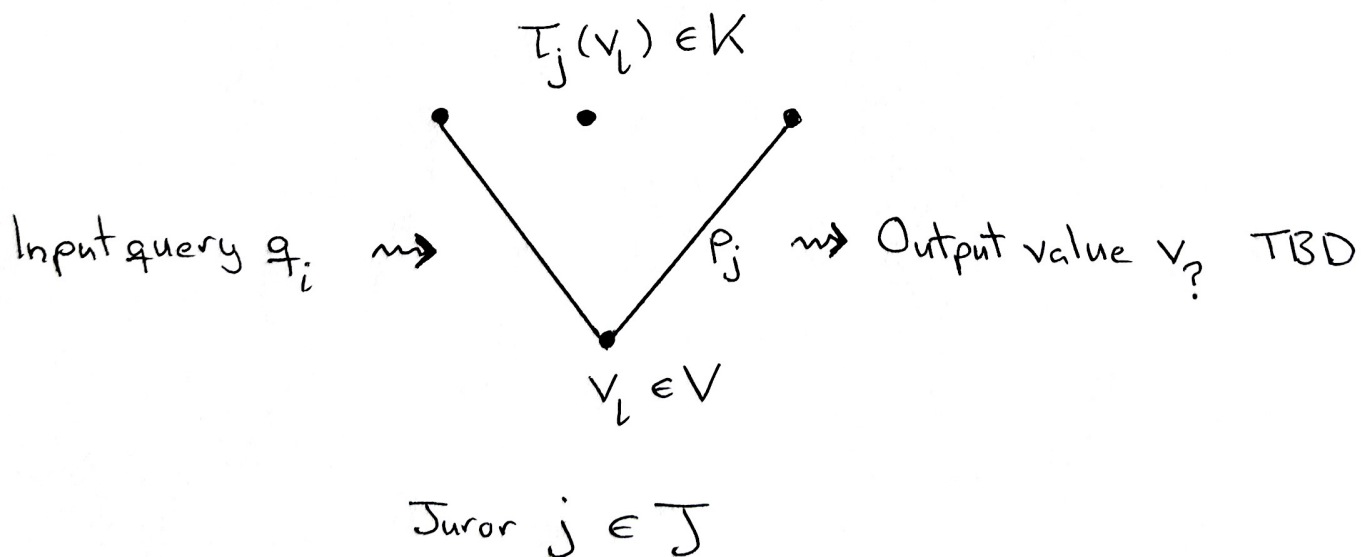
with $p_j(k_l^j) \approx_\varepsilon v_l$, $l \in [m]$.

$\mathcal{J}$ jurors j , each with their own key-value

semantics $p_j: K \rightarrow V$,

Let $\tau_j: V \rightarrow K$ be a section of p_j ,

that is $p_j(\tau_j(v_l)) \approx_\varepsilon v_l$, $l \in [m]$



Fix $i \in [n]$, we want to update and refine query q_i .

Let $q'_i \in Q'$ denote an update of q_i .

Let score be a query-key similarity function, like a reproducing kernel, and let $S(q, k)$ denote the probability distribution over keys after softmax normalization, given $q \in Q$.

Fix $j \in [J]$, define $l_j^* = \arg \max_l S(q'_i, \tau_j(v_l))$

and let $S_j \triangleq S(q'_i, \tau_j(v_{l_j^*}))$.

Define $j^* = \arg \max_j S_j$ and return output value

$V_{\text{out}} = v_{l_{j^*}^*}$, in answer to query q_i .

Refinement process:

- Replace q_i with q'_i
- Return updated value V_{out}
- Iterate until perplexity is acceptable

Note that this attention depends on q'_i and τ_j .

OK, so how do we find updates q'_i and sections τ_j ?

In a category with neural network arrows, consider

the following Joyal-Tierney symbolic calculus diagram:

$$\begin{array}{ccc}
 Q \otimes (\bigotimes_j K) & \xrightarrow{m \otimes (\bigotimes_j 1)} & Q' \otimes (\bigotimes_j K) \\
 \downarrow 1 \otimes (\bigotimes_j p_j) & \nearrow \sigma & \downarrow 1 \otimes (\bigotimes_j p_j) \\
 Q \otimes (\bigotimes_j V) & \xrightarrow{m \otimes (\bigotimes_j 1)} & Q' \otimes (\bigotimes_j V)
 \end{array} \quad (+)$$

Learn lifting neural network σ so that $(+)$ commutes, ie

$$i) \quad \sigma \begin{bmatrix} q_i \\ \vdots \\ p_j(k_l^j) \\ \vdots \end{bmatrix} =_{\varepsilon} \begin{bmatrix} m(q_i) \\ \vdots \\ k_l^j \\ \vdots \end{bmatrix}, \quad i \in [n], j \in [J], l \in [m]$$

$(J+1) \times 1$

$$ii) \quad \pi_i \sigma \begin{bmatrix} q_i \\ \vdots \\ v_l \\ \vdots \end{bmatrix} =_{\varepsilon} m(q_i), \quad i \in [n], l \in [m]$$

$$iii) \quad p_j \pi_j \sigma \begin{bmatrix} q_i \\ \vdots \\ v_l \\ \vdots \end{bmatrix} =_{\varepsilon} v_l, \quad j \in [J], i \in [n], l \in [m]$$

Fix $i \in [n]$, let $\iota : \bigotimes_j V \rightarrow Q \otimes (\bigotimes_j V)$

$$\bigotimes_j v \mapsto q_i \otimes (\bigotimes_j v)$$

and define $\tau_j : V \rightarrow K$ by

$$\tau_j(v) \triangleq \pi_j \circ \iota \left(\bigotimes_j v \right), \quad j \in [J],$$

then iii) says $p_j(\tau_j(v_l)) =_\epsilon v_l$, $l \in [m]$, $j \in [J]$,

that is τ_j is a section of p_j , $j \in [J]$.

—

We still have to determine update q'_i .

Given query q_i and juror j , let $\alpha_l^j = S(q_i, k_l^j)$

then $0 \leq \alpha_l^j \leq 1$ and $\sum_{l=1}^m \alpha_l^j = 1$.

Now define $\beta_{il} \triangleq \text{softmax}_l \left(\sum_{j=1}^J \alpha_l^j \right)$

and $q_i^0 \triangleq \sum_{l=1}^m \beta_{il} v_l$, an initial consensus update

by the jury.

Let $k_{l^*}^j = \arg \max_l \text{score}(q_i^0, k_l^j)$.

From $(\dagger)$, learn Fibration lifting σ

with $m(q_i) \triangleq_\varepsilon q_i^0$, and define final update

$$q_i^1 \triangleq \pi_i \sigma \left[\begin{array}{c} q_i^0 \\ \vdots \\ p_j(k_{i*}^j) \\ \vdots \end{array} \right]$$

Note that q_i^1 updates the consensus view q_i^0 while taking into account the different juror world views given by key-value semantics

$$p_j : K \rightarrow V, \quad j \in [J].$$

$q_i' \triangleq q_i^1$ is the update of query q_i we seek.

—

So now we have determined the updates q_i' and

sections T_j , using Fibration lifting σ from $(\dagger)$.

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Summary: We wish to update and refine a query q_i , $i \in [n]$, in a way that recognizes jurors have different key-value semantics $p_j : K \rightarrow V$.

Initially we use the p_j 's to create a consensus query q_i^0 acknowledging the jurors differing world views. With $m(q_i) =_\epsilon q_i^0$ in $(\dagger)$,

Fibration lifting σ learns how to update queries and at the same time take into account the differing semantics p_j of the jurors. This is the key step!

Finally we use the neural network σ that we just learned to refine q_i^0 into q_i^1 , our desired

update q_i' of q_i . $q_i' = q_i^1$ after iteration

is used to output value $V_{out} = V_{l_{j^*}^*}$, in answer

to query q_i .

Jury attention utilizes a fibration lifting to achieve consensus amongst differing views voiced by key-value semantics, a just transformer attention mechanism.



References

[1] 'Attention Legos'

Gary Nan Tie, Sep 8, 2025

DOI: 10.13140/RG.2.2.33398.05447

Attention building blocks for machines to learn world models?

[2] 'Deep Fibration Attention'

Gary Nan Tie, Aug 31, 2025

DOI: 10.13140/RG.2.2.17946.30402

In fibration attention, a lifting simultaneously updates a query and abductively finds a key to explain a value. This all depends on the key-value pairing (k,v) . By determining other keys that are highly similar to k , to also explain v , we can enhance key-value semantics to achieve more meaningful attention. Moreover, we can recursively layer keys upon keys in a tree, to deepen the set of value explanations.

[3] 'Contextual Fibration Attention'

Gary Nan Tie, Aug 29, 2025

DOI: 10.13140/RG.2.2.12205.35044

In a query-key-value attention model, the key-value pairing conveys language semantics. Query context also carries meaning. We introduce an abductive fibration attention model that holistically captures both. A fibration lifting, jointly learns from query context and language semantics (that abductively finds a key to best explain the returned value).

[4] 'Multi-head abductive attention'

Gary Nan Tie, Aug 25, 2025

DOI: 10.13140/RG.2.2.20597.23521

Transformer multi-head attention can focus on different parts of an input query sequence, however without regard to potentially different key-value semantics. This new primitive for formal query updating, constructs updates to a query in parallel, by learning multi-head abductive fibration liftings, to refine similarity based attention, utilizing the semantics of multiple key-value pairings.

[5] 'Fibformer - a fibration transformer'

Gary Nan Tie, Aug 22, 2025

DOI: 10.13140/RG.2.2.36293.10723

We introduce a new transformer with novel fibration attention based on abductive reasoning. This coherent approach to query updating enables controlled structured exploration of continuation values.